Craftax runs compared: achievements and what unlocks them

29 runs, 1.1M env steps each, size50m, all scored on the same 30 fixed worlds with tools/death_eval.py. Rows with several training seeds show seed 0. Updated 2026-11-07.

Every number is the mean episode-final achievement count over 30 evaluation episodes, ± its standard error over those worlds; unlock rates are the share of episodes in which an achievement fired; return is the mean raw Craftax return as a % of the maximum (226). An outside audit (2026-10-11, docs/audit-2026-10-11.md) found that earlier evaluations only shared the first world: later worlds depended on how long earlier episodes ran. All checkpoints were re-scored on fixed worlds, so every comparison below is paired world by world. The error bars cover the worlds only, not training seeds.

The same audit found three RSSM-2 bugs (non-moves counted as a step down, a new episode inheriting the old one's state, and training seeing up to 7 future steps). Honest map, B2, v1.2 and v2 were retrained with every fix ('fixed RSSM-2' rows). None improved: B2 8.27 vs 8.40, v2 7.17 vs 6.93, honest map 5.70 vs 6.17 -- all within noise -- and v1.2 fell from 9.97 to 8.77 (paired -1.20 ± 0.57), which one seed cannot tell apart from seed variation.

Three training seeds of each fixed run (mean over seeds, the seeds in brackets): B2 8.33 (8.27 / 7.70 / 9.03), v1.2 8.78 (8.77 / 8.83 / 8.73), v2 7.02 (7.17 / 7.83 / 6.07). Seeds of one arm differ by up to 1.8 achievements, more than most gaps between single runs. Comparisons below use the per-world mean over the three seeds.

What holds up on the fixed code: masking impossible actions (B2 vs honest map +2.63 ± 0.38, t = 7.0); v2 over vanilla DreamerV3 (+2.39 ± 0.28, t = 8.5); B2 over v2 (+1.31 ± 0.32, t = 4.1); the honest map over vanilla (+1.07 ± 0.24), though that run also has the potential shaping. What is not established: the manager over B2 (v1.2 vs B2 +0.44 ± 0.33, t = 1.3, though v1.2's three seeds agree within 0.1), the order among v1.1/v1.2/v1.3, and curiosity (v2 vs v2-cur +0.40 ± 0.39).

v2-taste (2026-11-01) removes v2's rule mask and adds taste over actions, a per-action bias learned from experience (docs/taste-prior.md). It scored 5.33 ± 0.19: -1.69 ± 0.25 below v2, level with the honest map (-0.37 ± 0.26) and above vanilla (+0.70 ± 0.28). It never made a wood pickaxe. Taste ended up disliking almost every special action, most of all the ones it pressed most: placing a table ended at -4.5 logits, as low as making an iron sword, which is impossible this early.

Normalized return, raw Craftax reward on the fixed worlds: best 4.04% (v1.2, old code); three-seed means 3.49% for v1.2 fixed, 3.31% for B2, 2.71% for v2 fixed; 1.96% for v2-taste and 1.65% for vanilla DreamerV3, against 7.09% for ITC (transformer world model, pixels). The B2 and v1.x runs read the game's action rules and hand-written goals; v2 keeps only the action rules; v2-taste reads neither.

Noise (2026-11-03, docs/ablations.md): rescoring seven checkpoints on a second set of 30 worlds moves one 30-world score by about ±0.55, more than its SE suggests. A gap under about 1.1 between two single runs is not evidence; 60 worlds and training seeds are.

What each addition does alone, on plain DreamerV3 with no mask and no shaping (2026-11-05): plain 4.77 (two seeds), + RSSM-2 5.33, + RSSM-2 + the v1.2 manager 5.53 (two seeds: 5.77 / 5.30), + taste over actions 4.63. The manager over plain is +0.77 ± 0.22, over RSSM-2 alone +0.20 ± 0.24, and level in training. The mask under the v1.2 manager: removing it drops v1.2 from 9.03 to 6.13 over 60 worlds (+2.90 ± 0.26), as much as without a manager.

The next agent (next-taste: fixed v2 with v1.2's named goals and taste on the manager) scores 9.38 ± 0.34 over 60 worlds (9.80 on these 30): +0.35 ± 0.35 over v1.2, +0.85 ± 0.39 over B2, +2.78 ± 0.48 over v2, with no tech-tree shaping. Taste locked its manager onto one goal (place table) at its first update.

The control, Next, no taste (the same agent with taste adding nothing), scores 11.30 ± 0.27 over 60 worlds (11.20 on these 30), the best of any run: +1.92 ± 0.41 over next-taste, +2.27 ± 0.33 over v1.2, +2.77 ± 0.32 over B2, and 4.56% normalized return. It makes the stone sword in 50% of episodes and the stone pickaxe in 30%. One seed.

Every run, side by side

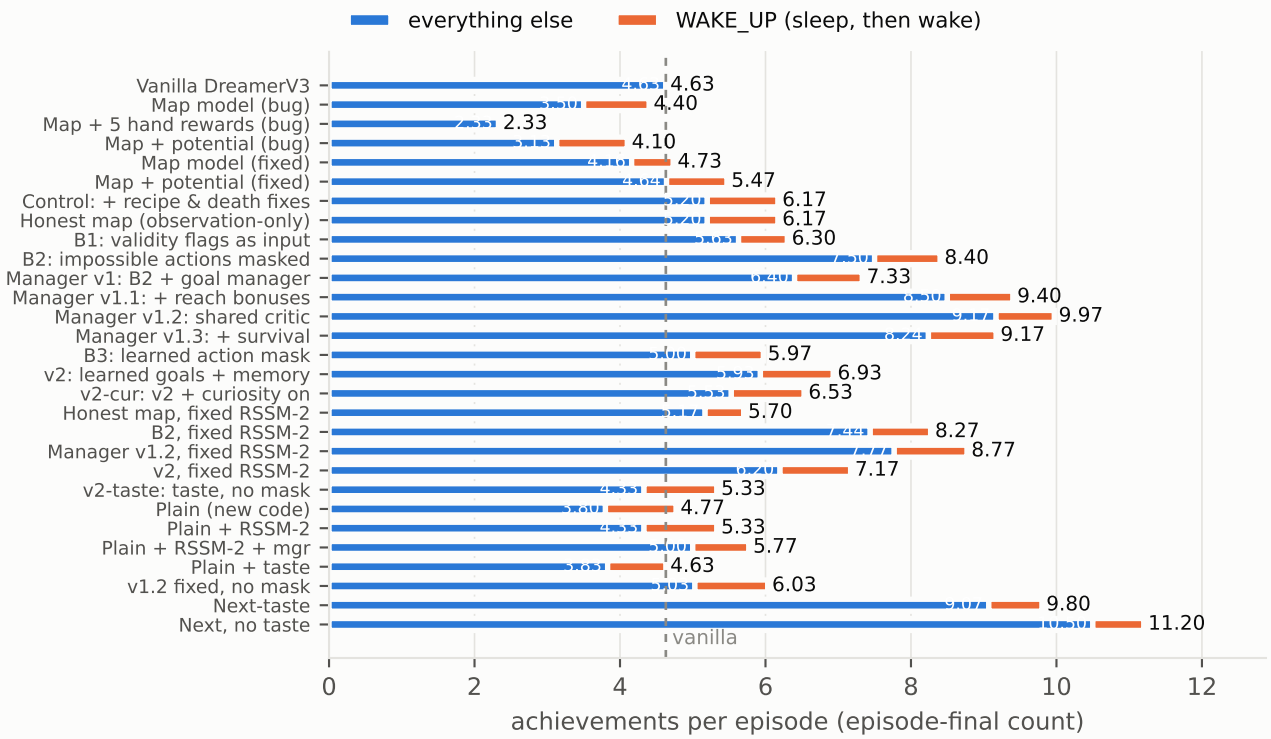
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run	ach. ± SE	w/o wake	life	return	table	pickaxe	sword	stone	drink
Vanilla DreamerV3	4.63 ± 0.22	4.63	295	1.65%	63%	23%	3%	0%	77%
Map model (bug)	4.40 ± 0.29	3.50	263	1.55%	40%	0%	0%	0%	47%
Map + 5 hand rewards (bug)	2.33 ± 0.19	2.33	265	0.63%	0%	0%	0%	0%	43%
Map + potential (bug)	4.10 ± 0.19	3.13	275	1.42%	17%	0%	0%	0%	43%
Map model (fixed)	4.73 ± 0.29	4.16	276	1.70%	67%	10%	0%	0%	77%
Map + potential (fixed)	5.47 ± 0.34	4.64	264	2.02%	70%	33%	13%	17%	60%
Control: + recipe & death fixes	6.17 ± 0.23	5.20	304	2.33%	97%	0%	50%	0%	80%
Honest map (observation-only)	6.17 ± 0.34	5.20	280	2.33%	73%	40%	17%	7%	77%
B1: validity flags as input	6.30 ± 0.33	5.63	295	2.39%	93%	53%	7%	27%	77%
B2: impossible actions masked	8.40 ± 0.54	7.50	282	3.32%	93%	73%	37%	60%	47%
Manager v1: B2 + goal manager	7.33 ± 0.53	6.40	256	2.85%	90%	57%	27%	37%	67%
Manager v1.1: + reach bonuses	9.40 ± 0.54	8.50	274	3.76%	93%	80%	43%	80%	90%
Manager v1.2: shared critic	9.97 ± 0.36	9.17	262	4.04%	100%	93%	50%	87%	67%
Manager v1.3: + survival	9.17 ± 0.49	8.24	294	3.66%	97%	80%	43%	67%	97%
B3: learned action mask	5.97 ± 0.28	5.00	269	2.24%	73%	33%	10%	7%	77%
v2: learned goals + memory	6.93 ± 0.36	5.93	270	2.67%	93%	37%	20%	27%	73%
v2-cur: v2 + curiosity on	6.53 ± 0.22	5.53	319	2.50%	100%	23%	30%	7%	80%
Honest map, fixed RSSM-2	5.70 ± 0.21	5.17	268	2.12%	87%	7%	50%	0%	77%
B2, fixed RSSM-2	8.27 ± 0.44	7.44	274	3.32%	97%	80%	40%	57%	67%
Manager v1.2, fixed RSSM-2	8.77 ± 0.52	7.77	319	3.48%	97%	67%	33%	67%	73%
v2, fixed RSSM-2	7.17 ± 0.44	6.20	276	2.77%	90%	53%	23%	30%	87%
v2-taste: taste, no mask	5.33 ± 0.19	4.33	280	1.96%	70%	0%	0%	0%	87%
Plain (new code)	4.77 ± 0.28	3.80	265	1.71%	50%	3%	10%	0%	73%
Plain + RSSM-2	5.33 ± 0.22	4.33	251	1.96%	80%	0%	27%	0%	77%
Plain + RSSM-2 + mgr	5.77 ± 0.24	5.00	255	2.15%	97%	37%	17%	3%	83%
Plain + taste	4.63 ± 0.29	3.83	262	1.65%	50%	0%	13%	0%	43%
v1.2 fixed, no mask	6.03 ± 0.33	5.03	275	2.27%	70%	33%	13%	20%	70%
Next-taste	9.80 ± 0.42	9.07	287	3.94%	100%	90%	60%	73%	93%
Next, no taste	**11.20 ± 0.41**	**10.50**	**284**	**4.56%**	**100%**	**100%**	**83%**	**97%**	**77%**

ach. ± SE = mean episode-final achievements ± standard error over the 30 worlds (one training seed, so seed variation is not included). w/o wake = the same minus WAKE_UP. life = mean lifespan in steps. return = raw Craftax return, % of 226. Rates are % of episodes. Bold rows are new in this report (2026-11-07). 'Fixed RSSM-2' rows were trained on the code with the RSSM-2 and shaping fixes of 2026-10-11; the older rows are the original checkpoints, re-scored. 'bug' = imag_shift True (fixed 2026-09-20). B1/B2: validity flags as an encoder input / impossible actions masked. mgr v1-v1.3: B2 plus the two-level agent (docs/design-manager.md). B3: the mask learned instead of read from the rules. v2: learned goals, curiosity, RSSM-2 as a memory, rule mask kept (docs/design-v2.md); v2-cur: v2 with curiosity actually scaled. v2-taste: fixed v2 with the rule mask and the validity flags removed and taste over actions added (docs/taste-prior.md). Plain / + RSSM-2 / + mgr / + taste: plain DreamerV3 on the 2026-11 code with one addition each, no mask, no shaping (docs/ablations.md). v1.2 fixed, no mask: fixed v1.2 minus the rule mask and the validity flags. Next-taste / Next, no taste: fixed v2 with the 13 named goals of v1.2 in place of learned goals, with and without taste on the manager (docs/taste-prior.md; the control is taste's lam_mgr 0). Rows with several seeds show seed 0; the seed means are in the summary.

Sleep inflates the map runs; the mask is the large gain

Mean episode-final achievements per run, split into WAKE_UP and the rest; unlock rate of the early tech-tree rungs

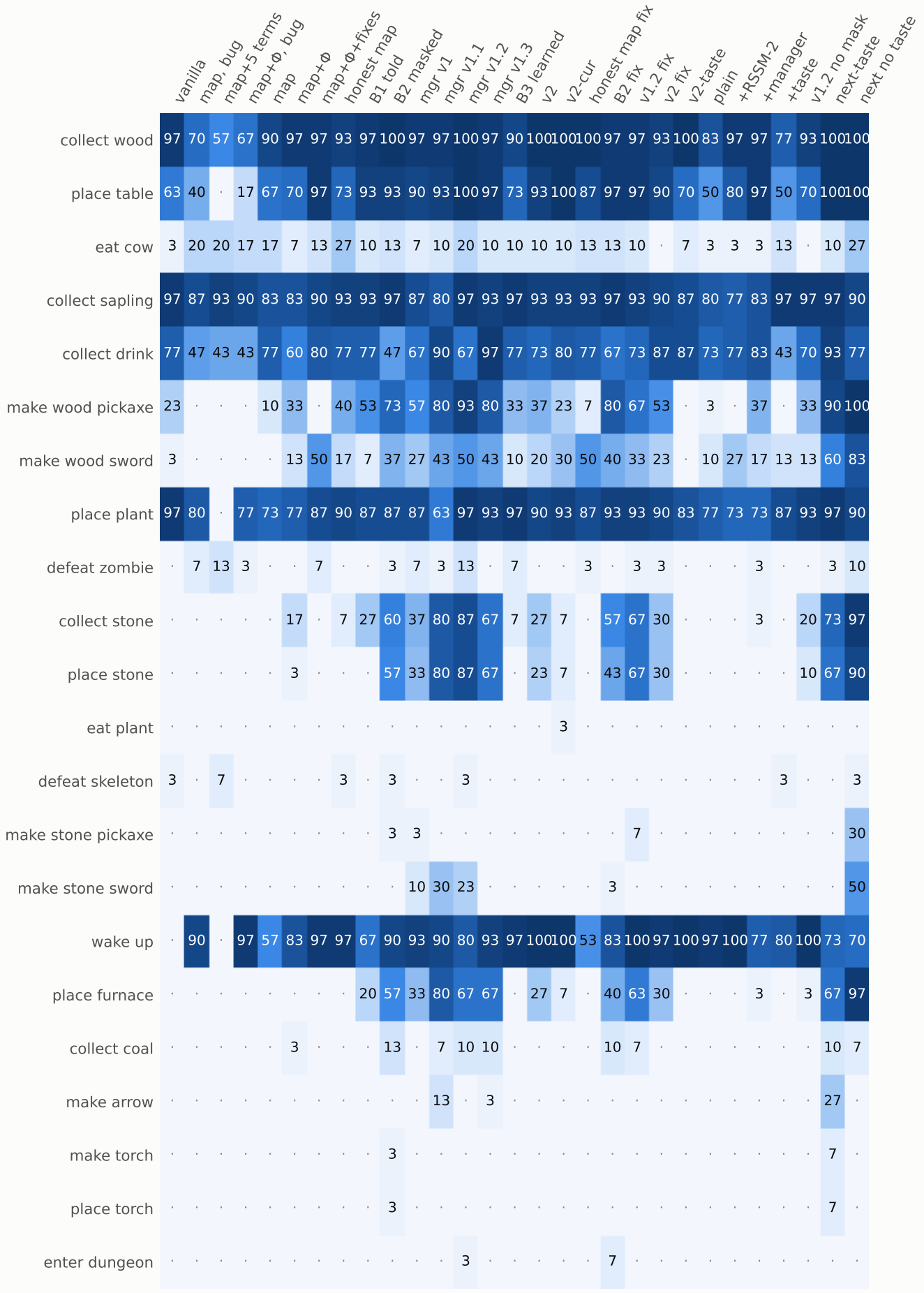


Unlock rate (% of episodes), early tech tree in dependency order

	collect wood	place table	make wood pickaxe	make wood sword	collect stone	place stone	place furnace	collect coal	make stone pickaxe
Vanilla DreamerV3	97	63	23	3	0	0	0	0	0
Map model (bug)	70	40	0	0	0	0	0	0	0
Map + 5 hand rewards (bug)	57	0	0	0	0	0	0	0	0
Map + potential (bug)	67	17	0	0	0	0	0	0	0
Map model (fixed)	90	67	10	0	0	0	0	0	0
Map + potential (fixed)	97	70	33	13	17	3	0	3	0
Control: + recipe & death fixes	97	97	0	50	0	0	0	0	0
Honest map (observation-only)	93	73	40	17	7	0	0	0	0
B1: validity flags as input	97	93	53	7	27	0	20	0	0
B2: impossible actions masked	100	93	73	37	60	57	57	13	3
Manager v1: B2 + goal manager	97	90	57	27	37	33	33	0	3
Manager v1.1: + reach bonuses	97	93	80	43	80	80	80	7	0
Manager v1.2: shared critic	100	100	93	50	87	87	67	10	0
Manager v1.3: + survival	97	97	80	43	67	67	67	10	0
B3: learned action mask	90	73	33	10	7	0	0	0	0
v2: learned goals + memory	100	93	37	20	27	23	27	0	0
v2-cur: v2 + curiosity on	100	100	23	30	7	7	7	0	0
Honest map, fixed RSSM-2	100	87	7	50	0	0	0	0	0
B2, fixed RSSM-2	97	97	80	40	57	43	40	10	0
Manager v1.2, fixed RSSM-2	97	97	67	33	67	67	63	7	7
v2, fixed RSSM-2	93	90	53	23	30	30	30	0	0
v2-taste: taste, no mask	100	70	0	0	0	0	0	0	0
Plain (new code)	83	50	3	10	0	0	0	0	0
Plain + RSSM-2	97	80	0	27	0	0	0	0	0
Plain + RSSM-2 + mgr	97	97	37	17	3	0	3	0	0
Plain + taste	77	50	0	13	0	0	0	0	0
v1.2 fixed, no mask	93	70	33	13	20	10	3	0	0
Next-taste	100	100	90	60	73	67	67	10	0
Next, no taste	100	100	100	83	97	90	97	7	30

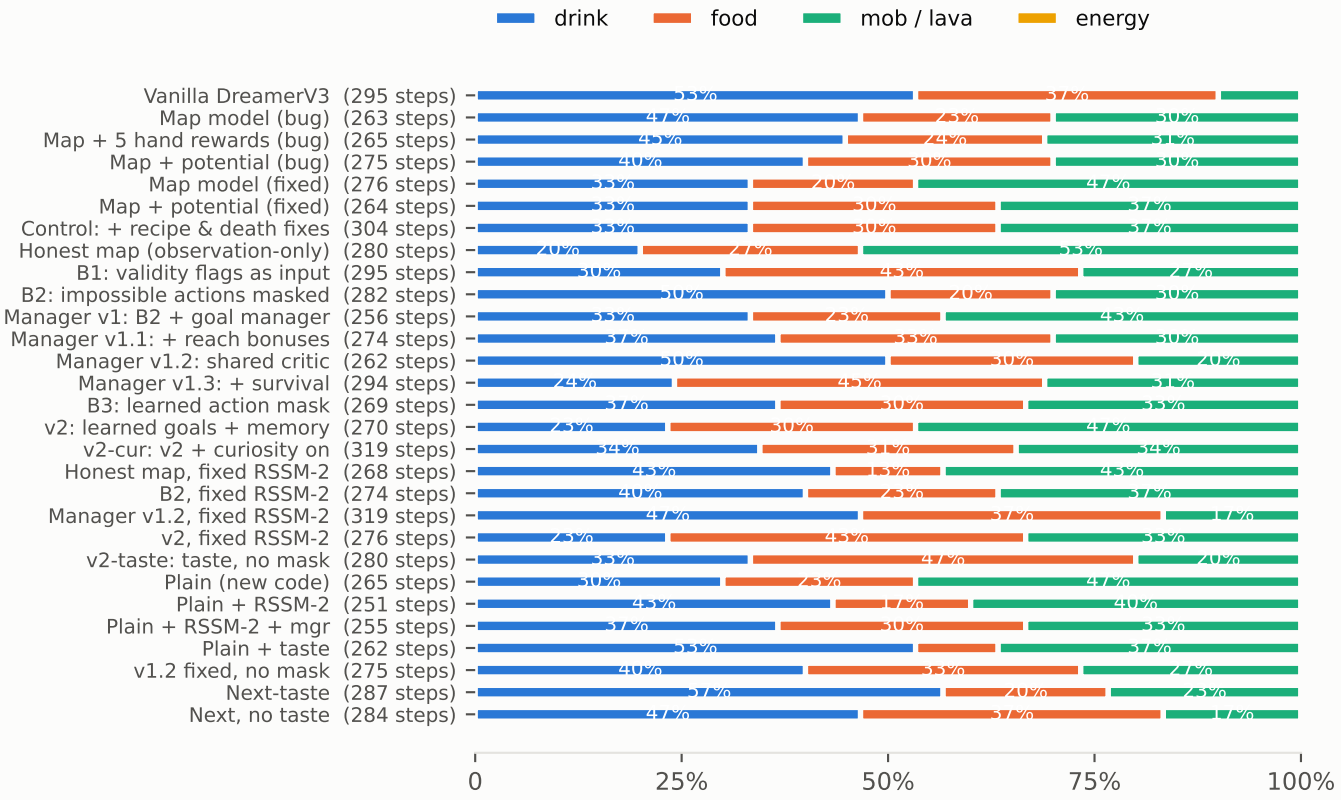
Every achievement any run unlocked

% of evaluation episodes in which each achievement fired. The other 45 of 67 were never unlocked by any run.



Survival never moved

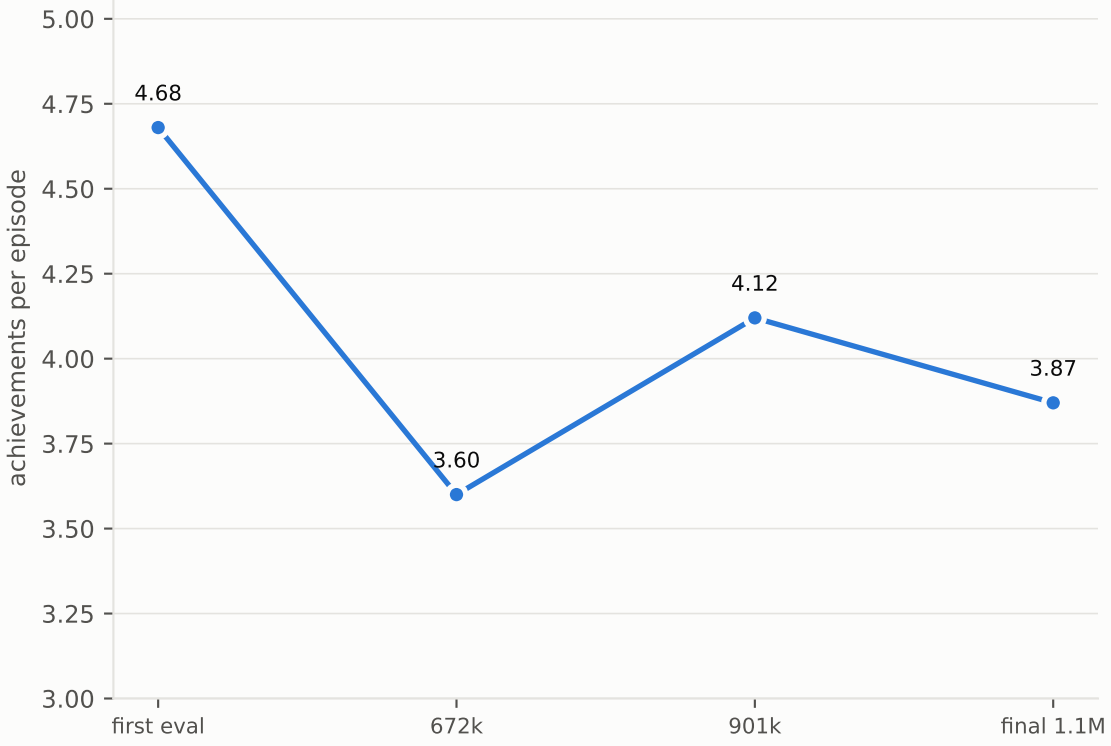
Cause of death (share of evaluation episodes) and lifespan; and how much one run moves between its own checkpoints



Random policy lives 261-284 steps; doing nothing lives 333. Map + potential (bug) has no recorded causes.

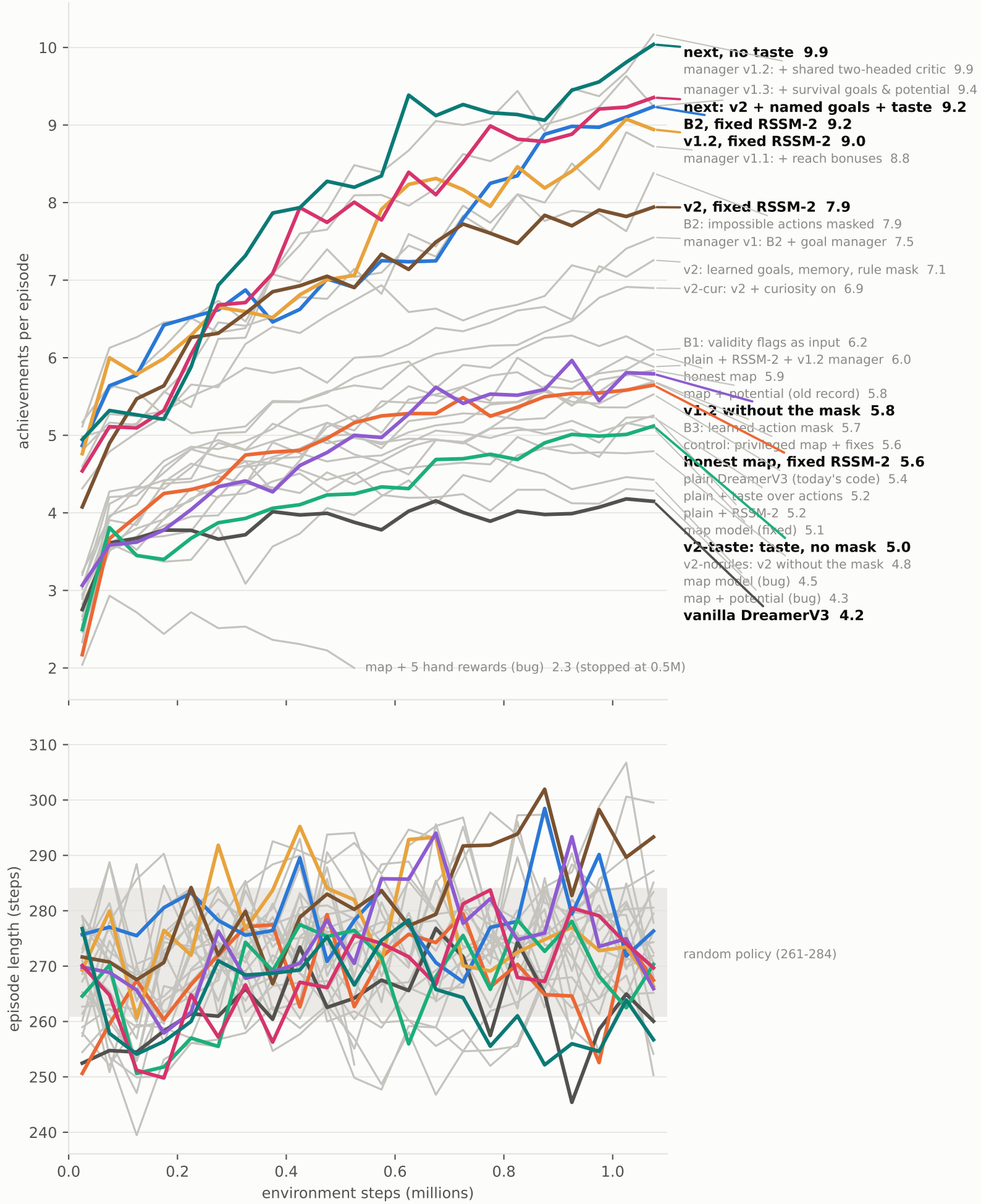
One run, four checkpoints: Map + potential (bug)

The same training run, scored at four points (25-30 episodes each), moves by 1.08 achievements. That is the scale of noise any comparison between single runs has to beat.



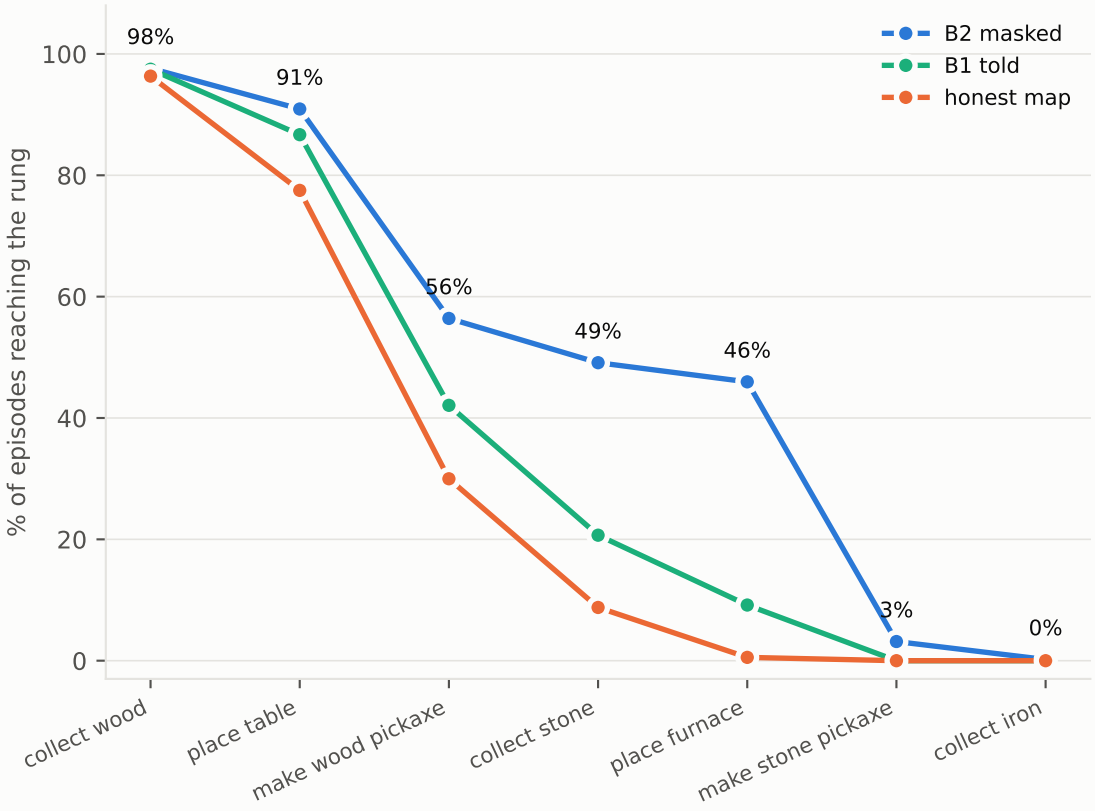
Training curves: every run

Rebuilt from each run's replay; 50k-step bins; value = mean of the last 100k steps; one seed per run.



Where B2 falls short

Last 150k training steps of each run (~500 episodes), decoded from replay by tools/diagnose_run.py



1. The stone pickaxe is the wall. B2 reaches the furnace in 46% of episodes but makes the stone pickaxe in 3%. Of the 249 episodes that collected stone, 73% held wood and stone at the same time, but only 22% ever stood next to a table while holding both, and 6% made the pickaxe. Two gaps: it rarely brings the ingredients back to a table (logistics), and when it does the chance lasts one step -- the single log usually goes into the wood sword or it walks off (measured in docs/b2-casebook.pdf).
2. Table to wood pickaxe leaks. 38% of the episodes that place a table never make the wood pickaxe. At a table holding wood it crafts every time, but every one of these episodes placed the table with its last two logs, and later logs go into another table rather than back to the first (docs/b2-casebook.pdf).
3. Survival. Training-window lifespan 285 (median 275) against 261-284 for a random policy. Thirst is still the first meter to run out in 45% of deaths and hunger in 24%: Craftax pays for the first drink only, so topping up is never rewarded. A hostile mob is adjacent at 61% of deaths; it dies in lava in 4% (the others: under 1%), the price of exploring further. Deaths in its sleep are rare (5%).
4. Wasted presses. 43% of presses still do nothing (experiment A): DO with nothing in front (26%), NOOP and walking into walls. Masking only removes the special actions; these basic ones are never masked.

What the results say, and what would settle the rest

0. Masking impossible actions is the one large, robust gain

B2 beats the same model without the mask by +2.63 ± 0.38 achievements on the fixed code (three-seed mean; +2.23 on the old), on the same 30 worlds. Buttons that were never pressed where they would work (the wood pickaxe, the furnace, placing stone) get pressed once a useless press can no longer teach the agent that the button is bad. The mask reads the game's rules, so it is a separate arm from the published, knowledge-free agents. It holds under the manager too: v1.2 without the mask scores 6.13 over 60 worlds against 9.03 (+2.90 ± 0.26), and +2.8 in training.

1. The manager's lead is not established

Over 60 worlds v1.2 leads B2 by +0.50 ± 0.25, but the two are level in training (8.75 vs 8.81 over the last 100k steps). Without the mask the manager adds little: v1.2 without it is +0.33 ± 0.40 over the honest map. On plain DreamerV3, a second seed did not repeat the manager's first seed (5.30 vs 5.77; it ended training below plain): over RSSM-2 alone it is +0.20 ± 0.24. Old v1.2's 9.97 was 9.73 over 60 worlds: a real lead, possibly from the pre-fix RSSM-2 leak, plus some world luck.

2. v2 clearly beats vanilla DreamerV3 -- but not B2

v2 drops the hand-written goals, the reward potentials and the map labels, and still beats vanilla by +2.39 ± 0.28 (fixed code, three seeds). It stays at the wood tier: no stone tool in any episode. B2 leads it by +1.31 ± 0.32. Curiosity, once scaled (v2-cur), changed nothing measurable.

3. The RSSM-2 fixes did not explain the results

Training no longer sees future steps, episodes no longer leak, and non-moves no longer count as a step down. The fixed runs land within noise of the old ones (v1.2 aside). The map model's contribution was real but not driven by the leak.

4. Survival is still unsolved

Lifespans stay near the random band (255-320 steps); thirst, hunger and mobs still kill most episodes. The longest-living fixed run is v1.2 (319 steps), which crafts less than before.

5. Taste did not work, over actions or on the manager

v2-taste removed v2's rule mask and learned a bias for each special action: 5.33, below v2 (-1.69) and level with its no-taste control. On plain DreamerV3, centered taste over actions did nothing (4.63 vs 4.77). On the manager (next-taste), taste picked one goal, place table, at its first update (6.7k steps) and held it to the end (goal entropy 0): only a chosen goal gets evidence, and centering over the chosen goals leaves the winner's evidence near zero, so nothing moves it back. Taste outweighed the critic's spread between goals 15 to 300 times over. Against its control, taste cost -1.92 ± 0.41 over 60 worlds. It cut the manager off from every goal past the table: the actor was told "be near a table" 40% of the time and given no goal the other 60%, placed more tables and never made a stone tool.

6. The next agent without taste is the best run so far

Next, no taste scores 11.30 over 60 worlds and 9.92 in training, ahead of every fixed-code run at every checkpoint from 700k steps, without tech-tree shaping. It is fixed v2 with the 13 named goals in place of learned ones, and a manager that chooses by game value alone: it proposes all eleven goals, the stone tools and iron included. The gain over v2 (+4.18) comes from the named goals and a manager free to use them. One seed so far.

7. What would settle the rest

Two more seeds of Next, no taste. Then which part matters: the named goals, curiosity, or RSSM-2 as a memory. If taste is kept, it must sit on the critic's scale, reach goals the manager did not choose, and treat "no goal" like the others. The mask is still worth +2.6 to +2.9 wherever it was measured; a knowledge-free mask still needs to tell "does nothing" from "not yet learned".